

Objectives

- To evaluate the millions of association rules generated by the FP-Growth algorithm and isolate statistically significant economic patterns.
- To interpret Machine Learning metrics (Support, Confidence, Lift) to validate the strength of agricultural supply chain bundles and inflationary clusters.
- To design and generate comprehensive data visualizations that translate complex rule networks into accessible intelligence for policy analysis.

Tasks Completed Data Analyst & Visualizer (Rowena Bornilla)

- **Tasks:** Rule evaluation, metric analysis, and data visualization generation.
- **Description:**
 - Analyzed the massive outputs from the FP-Growth algorithm, utilizing strict filtering on the **Lift** metric to separate random item co-occurrences from strong, predictive economic relationships.
 - Developed **Support-Confidence-Lift bubble plots** using `matplotlib` and `seaborn` to visualize rule intensity and frequency across both transaction types.
 - Engineered **Directed Network Graphs** using `networkx` to map the directional flow of price spikes across commodities, applying the `kamada_kawai_layout` for optimized spatial rendering.
 - Created a **Regional Price Spike Heatmap** to identify specific agricultural vulnerabilities across Philippine LGUs by plotting spike frequencies of the top 20 commodities.
- **Screenshot/Code Snippet:**

Python

```
# Generating the Regional Price Spike Heatmap for Vulnerability Analysis
```

```
heat_data = (  
    spike_counts[spike_counts['item'].isin(top_items)]  
    .pivot_table(index='adm1_name', columns='item',  
                 values='spike_count', fill_value=0))
```

```
fig, ax = plt.subplots(figsize=(16, 7))
```

```
sns.heatmap(  
    heat_data,  
    cmap='YlOrRd',  
    linewidths=0,
```

```
annot=False,  
  
ax=ax,  
  
cbar_kws={'label': 'Spike Count'})  
  
ax.set_title('Regional Price Spike Frequency — Top 20 Commodities')
```

Tasks to be Completed

- Refine the threshold parameters on the Network Graphs to better visually separate disconnected commodity clusters, making the final export cleaner for the research manuscript.
- Synthesize the visual findings into a comprehensive "Economic Utility" summary detailing how LGUs can use the Heatmap for targeted food security interventions.

Results and Analysis

1. **Validation of Inflationary Contagion:** Through metric evaluation, I validated severe market contagion effects. The data confirmed that when core proteins like "Eggs" and "Meat (pork)" spike simultaneously, there is a 67.4% confidence (with a massive 6.56 Lift) that "Fish (milkfish)" and "Rice (regular, milled)" are also spiking. This visual and statistical proof shows how protein shortages immediately transfer pricing pressure to alternative staples.
2. **Visualizing Market Behavior:** The Bubble Plots successfully illustrated the density of the rules. The visualization proved that while Co-Occurrence rules are highly frequent (clustered at lower lift), the Price Spike rules, though fewer, exhibit significantly higher Lift intensity, marking them as highly reliable predictive indicators.
3. **Regional Vulnerability Mapping:** The Heatmap visualization provided immediate, actionable intelligence. It clearly highlighted regions with structurally weak supply chains that experience high frequencies of localized agricultural inflation, particularly in the vegetable sector, providing a direct map for resource allocation.

Challenges and Solutions

- **Challenge:** Rendering the Directed Network Graphs originally caused massive performance slowdowns (taking minutes to render) due to the sheer number of overlapping edges and complex connection styles native to `networkx`.
- **Solution:** I optimized the visualization code by removing complex, curved connection styles (forcing straight arrows which render in milliseconds) and utilized the `kamada_kawai_layout` over the standard force simulation. This resolved the rendering lag and produced clean, highly readable mapping of the price spike flow.

Conclusion This final analytical phase successfully bridged the gap between complex algorithmic outputs and real-world economic application. By critically evaluating the metrics and generating optimized, high-fidelity visualizations, I was able to translate millions of raw machine learning rules into concrete, verifiable insights. The visualizations—particularly the rule networks and the spike heatmap—provide a clear, data-driven foundation for addressing regional food insecurity and price instability across the Philippines.

